



Kabul Polytechnic University
Computer Science Faculty
Information System Department

Database II

Lecture I: Entity Relationship (ER) Model

17 August – 2026

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- ▶ Entity Type Classification
- ▶ Attribute Types
- ▶ Identifiers (Keys)
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Entity Relationship (ER) Model

E-R Model

- A **logical representation** of the data for an organization using:
 - **Entities** for categories of data and,
 - **Relationships** for associations between entities.

Elements of ER Model

■ Entity

- Person, place, object, event or concept for which data is collected and maintained. (**A table**)

■ Attribute

- The Property or characteristic of an entity. (**A field in a table**)

■ Relationship

- A logical connection between different entities.

Note that:

- **Entity Type/Class:** A set of related attributes.
- **Entity Instance:** is the member of entity type.
- **Entity Set:** is a group of entity instances

What should an Entity Be?

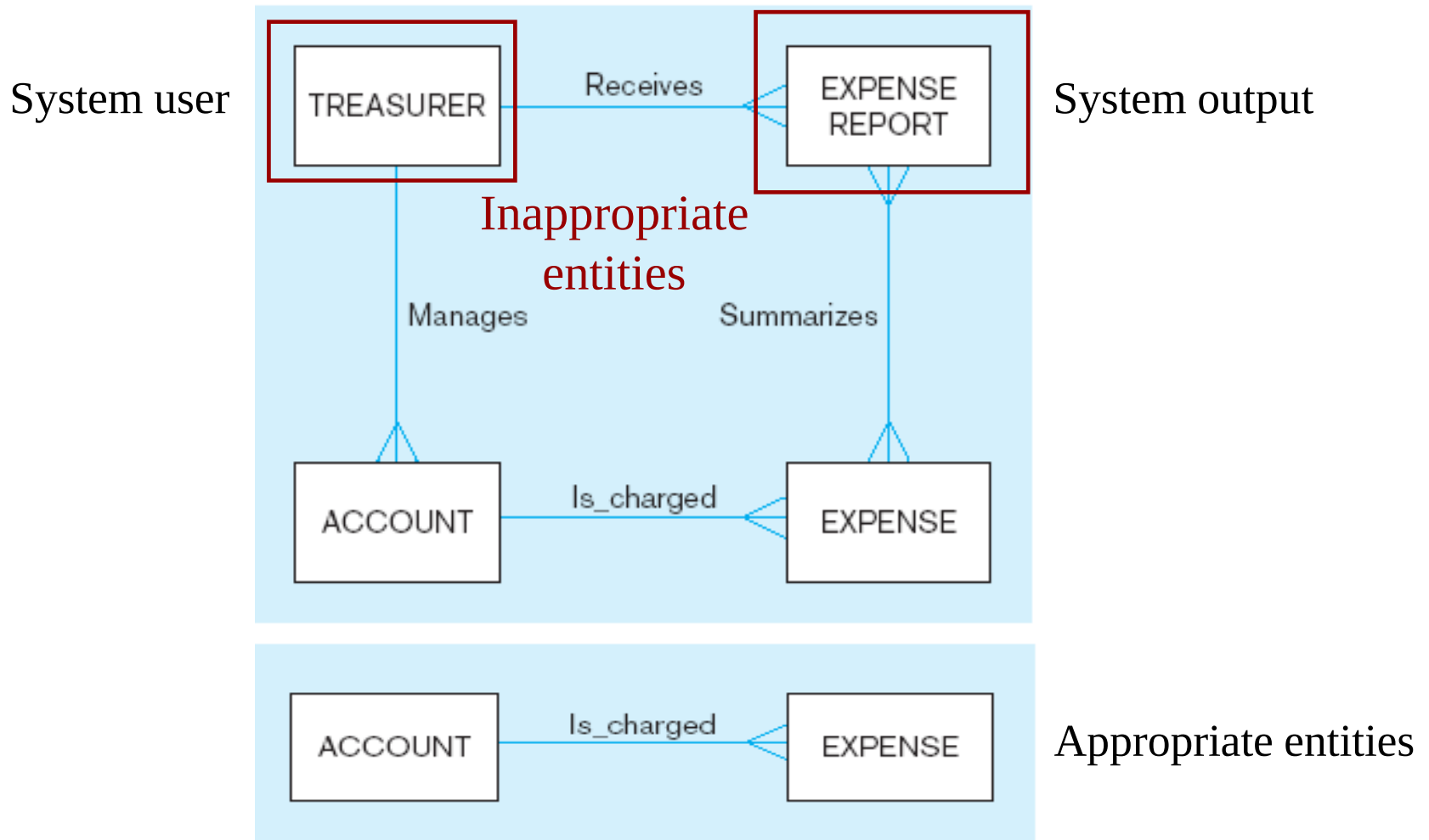
SHOULD BE:

- ✓ An object that will have many instances in the database.
- ✓ An object that will be composed of multiple attributes.
- ✓ An object that we are trying to model.

SHOULD NOT BE:

- A user of the database system.
- An output of the database system (e.g., a report).

Inappropriate Entities



Entity Type Classification

- **Strong Entity Type:**

- An entity that exists independently of other entity types.
- Also called Independent Entity Type.

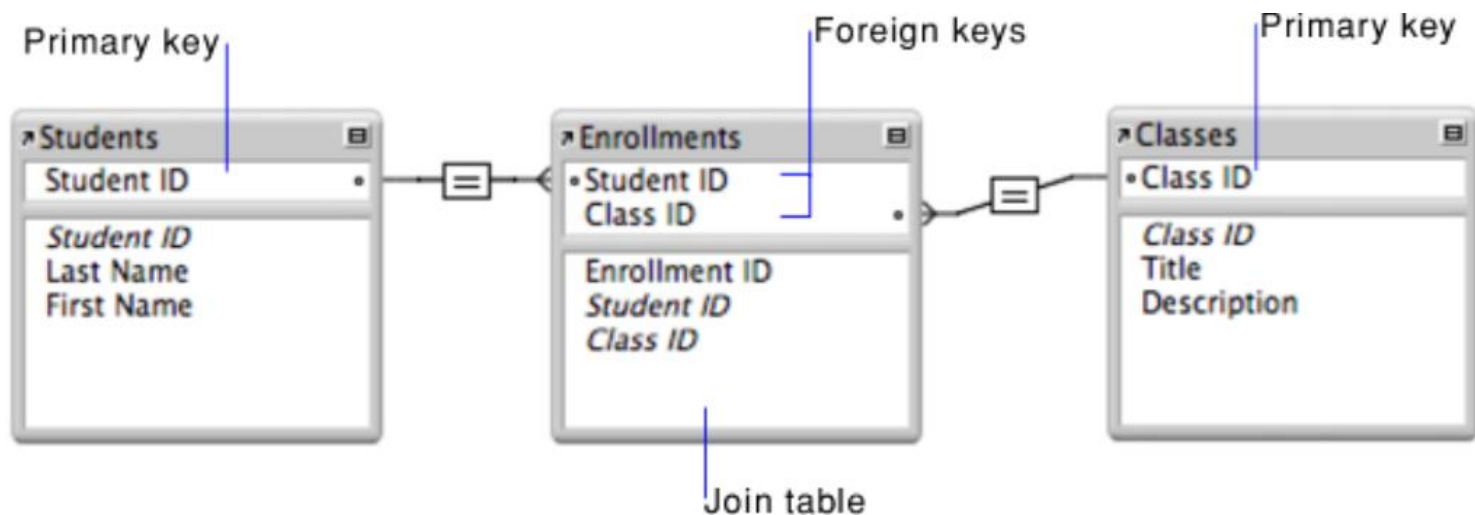
- **Weak Entity Type:**

- An entity type whose existence depends on some other entity type.
- Also called Dependent Entity Type.
- May have no key attribute.
- **For example:** RESERVATION is strong but RESERVATION_DETAILS is weak Entity Type.

Entity Type Classification

- **Associative Entity Type:**

- An entity type that **associates the instances** of one or more entity types with one another.



Attribute Types (1)

Attribute: Property or characteristic of an entity.

- **Simple (Atomic) Attribute:**

- Cannot be divided into smaller components.
- A single atomic value for the attribute.
- For example, **first name**, **month**, **ID**.

- **Composite Attribute:**

- Can be divided into smaller subparts, which represent more basic attributes with independent meanings.
- For example, **Address** or **Name**.

Attribute Types (2)

■ **Multi Valued Attribute:**

- An entity may have multiple values for that attribute.
- For example, **Skills of an employee, favorite colors or subjects.**

■ **Derived Attribute:**

- An attribute whose values can be calculated from related attribute values.
- For example, **Age** is derived from date of **birth of an employee.**

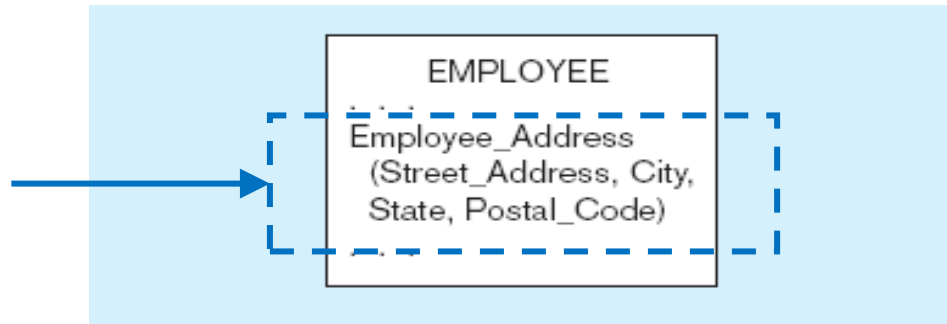
■ **Identifier Attribute:**

- An attribute whose value separates instances of an entity type.
- For example, **Employee_ID.**

Attribute Types (3)

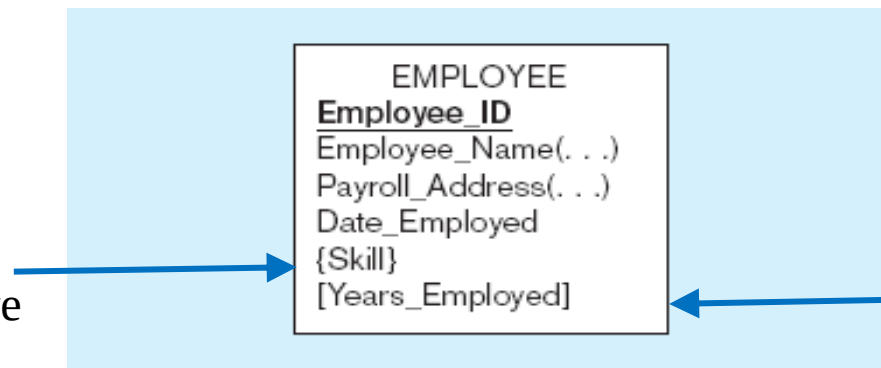
Composite Attribute:

An attribute broken into component parts.



Entity with **multivalued** attribute (Skill) and **derived** attribute (Years_Employed)

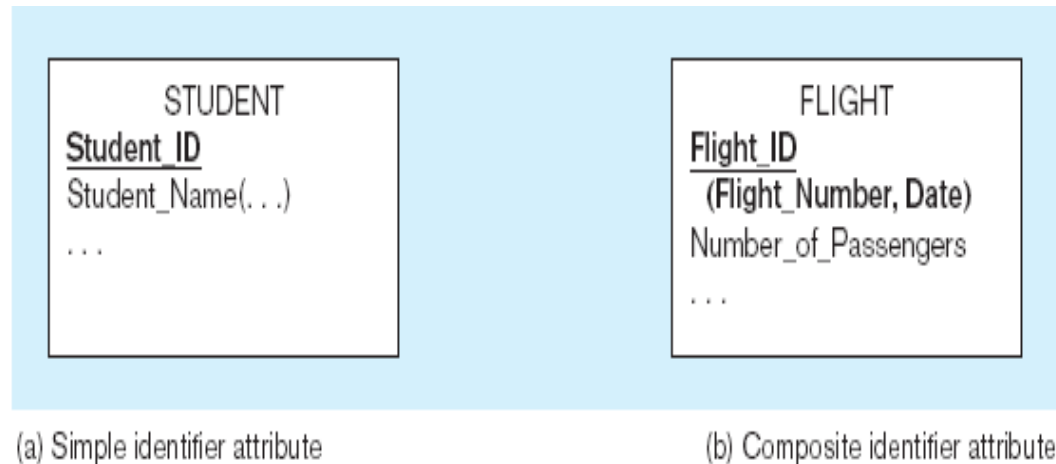
Multivalued
an employee can have
more than one skill.



Derived
from date employed
and current date

Identifiers (Keys)

- **Identifier (Key):** An attribute (or combination of attributes) that uniquely identifies individual instances of an entity type.
- **Simple** versus **Composite** Identifier.



- **Candidate Identifier:** An attribute that could be a key satisfies the requirements for being an identifier.

Characteristics of Identifiers

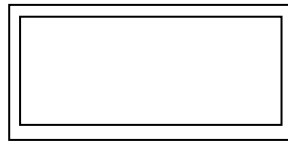
- i. Will not change in value
- ii. Will not be null
- iii. No intelligent identifiers
(e.g., containing locations or personal details that might change)
- iv. Simple keys, composite keys

Basic ER Notations

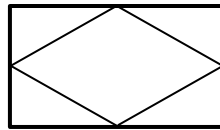
- **ER-D** is a graphical presentation of an entity-relationship model.
- **ER Notations:**



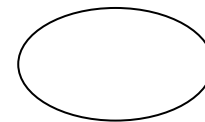
Entity Type



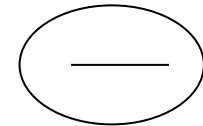
Weak Entity



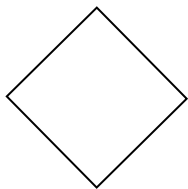
Associative Entity



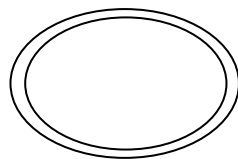
Attribute



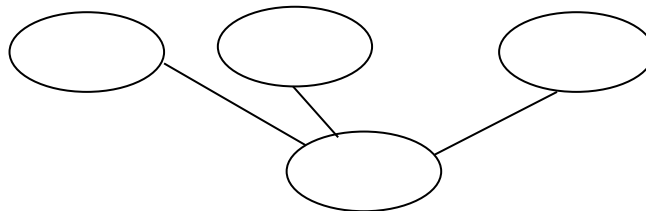
Key Attribute



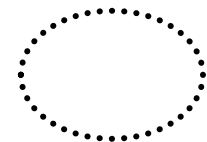
Relationship



Multi-valued Attribute

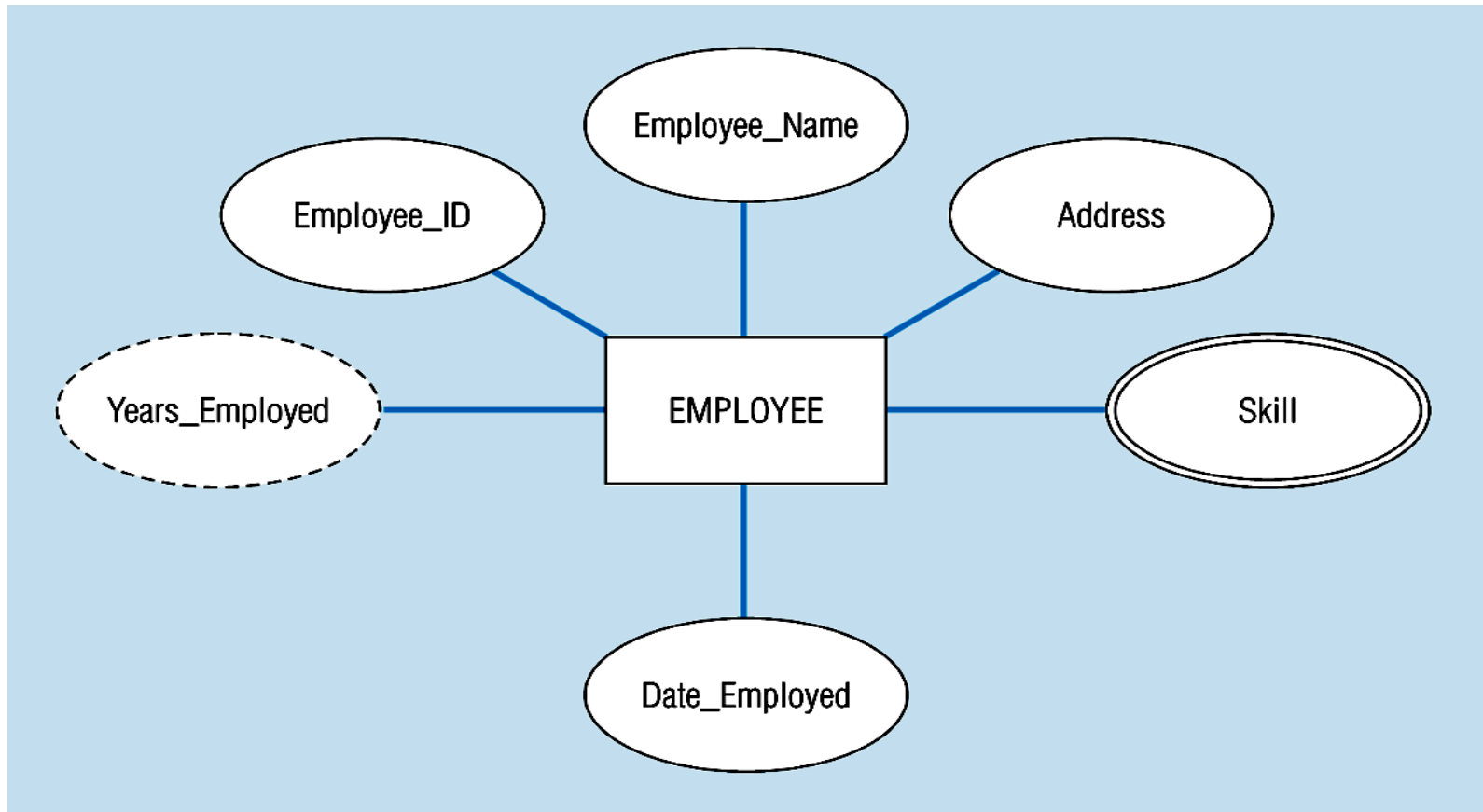


Composite Attribute

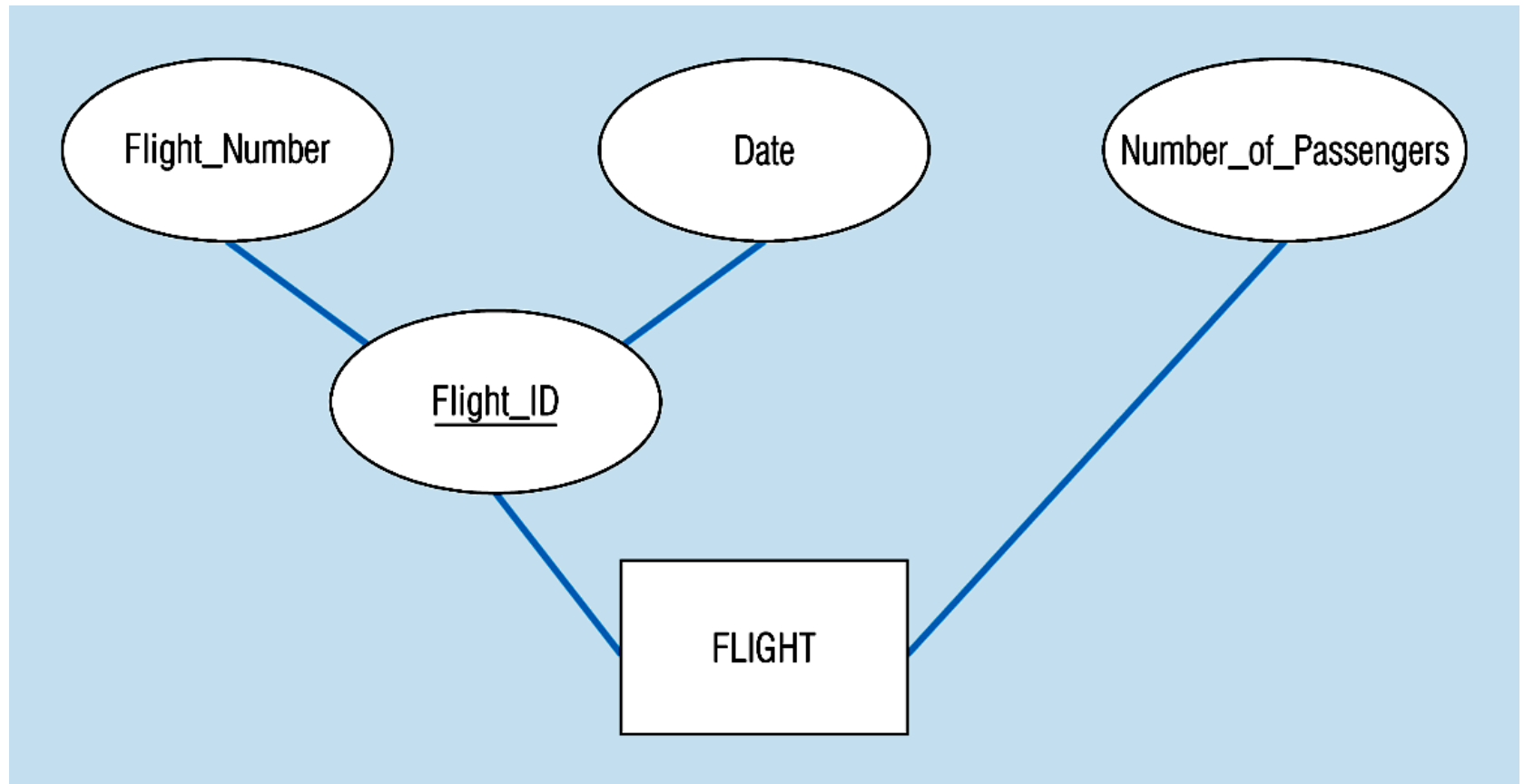


Derived Attribute

Example1: Entity Type & Attributes



Example2: Entity Type & Attributes



Relationship

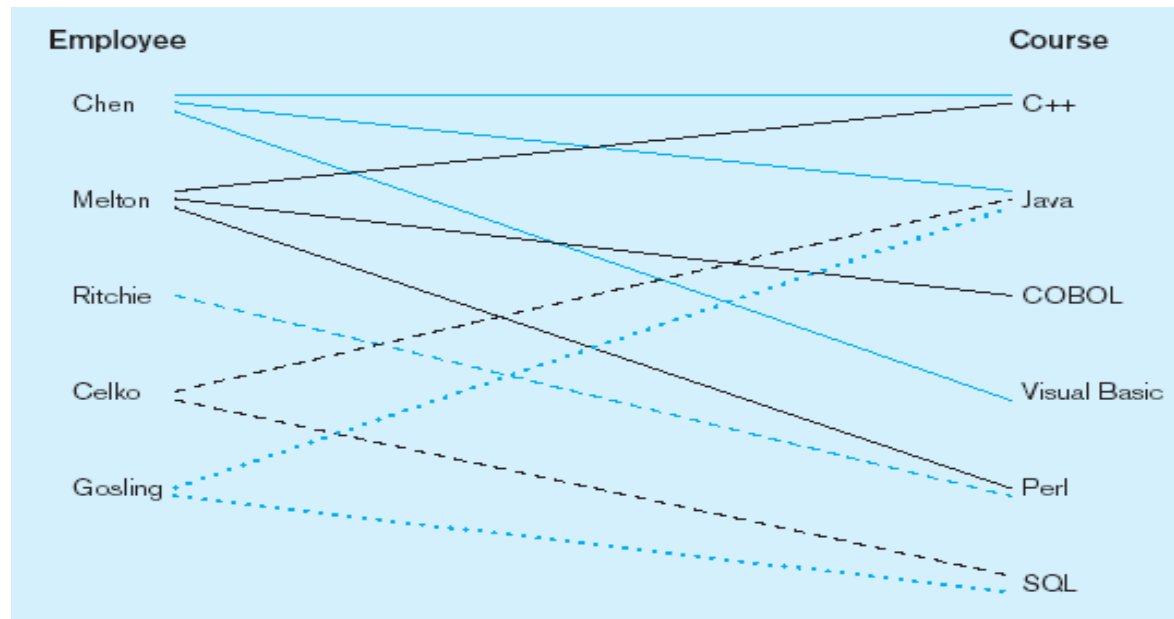
- An association between two or more entities types with a specific meaning.
 - For example, the **EMPLOYEE Khan** works on the **Education PROJECT**.
 - A relationship has a **name** and **degree** for confirmation.
 - Relationship is represented by:
 - **Diamond-shaped box** to display a relationship type.
 - **Straight lines** connected to the participating entity types.
 - The entities having relationship are called **participating entities**.

Relationship

a) Relationship type



b) Relationship instances



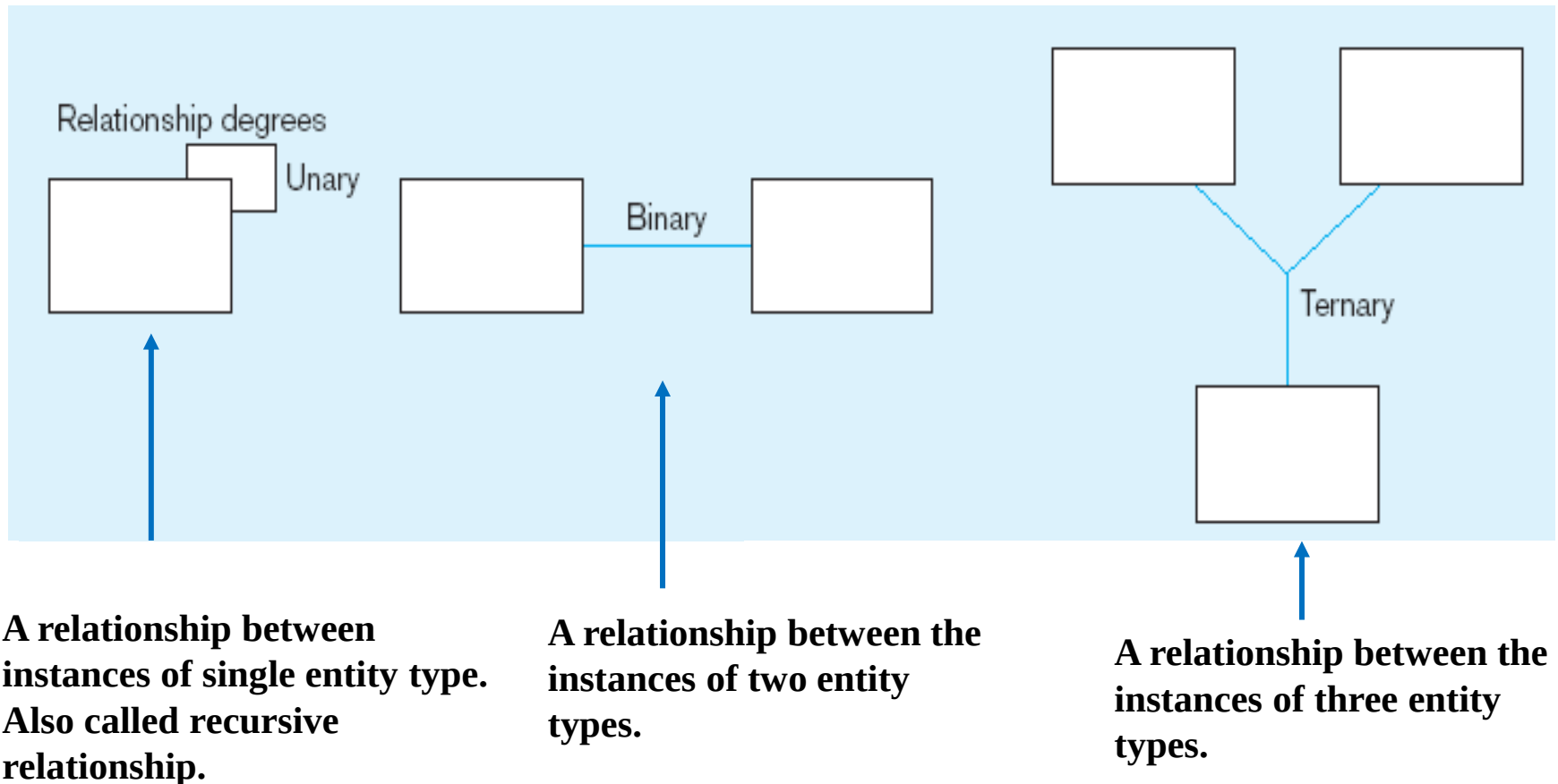
Degree of Relationship (1)

- The number of entity types that participate in a relationship.

Types of Relationships:

- ✓ **Unary Relationship**
- ✓ **Binary Relationship**
- ✓ **Ternary Relationship**

Degree of Relationship (2)



Cardinality of Relationship

- The maximum number of relations between entities.

- ✓ **One – to – One (1:1)**

- Both entities of the relationship can have one related instance. (person – passport, country – capital city)

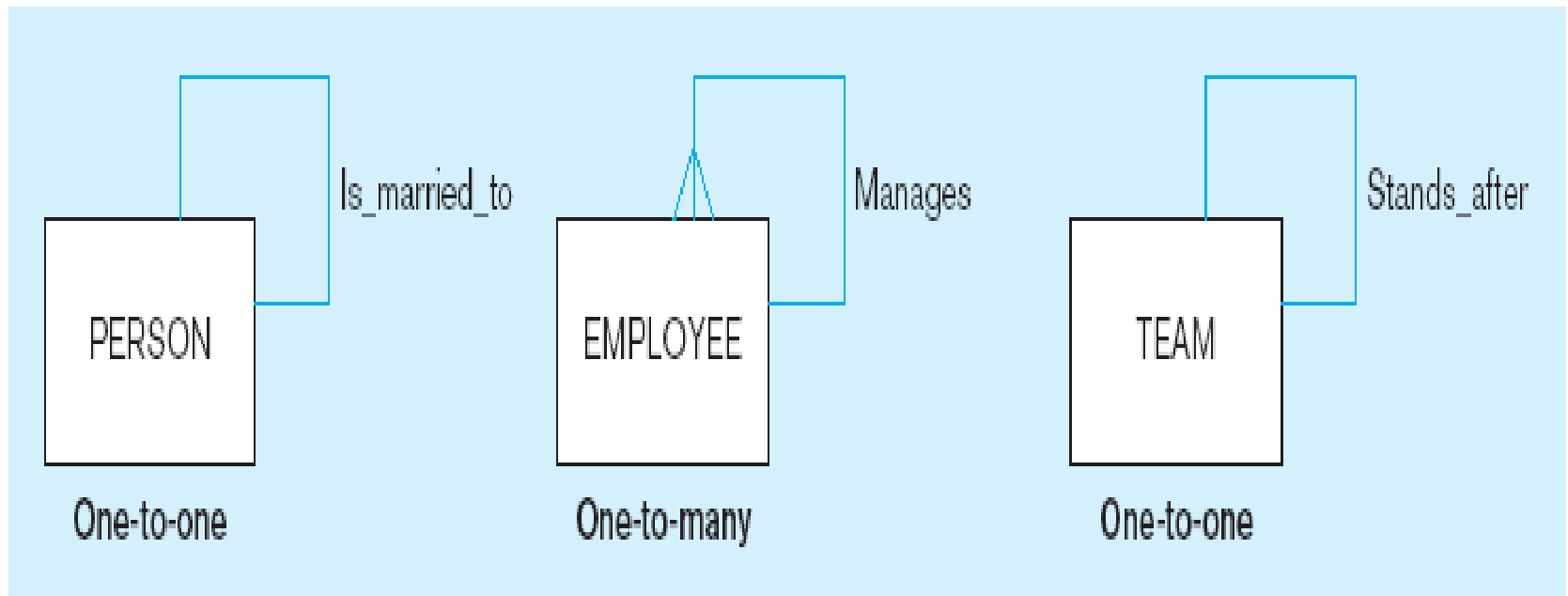
- ✓ **One – to – Many (1:N)**

- An entity can have many related instances, but the other entity will have a maximum of one related instance. (teacher – courses, department - student)

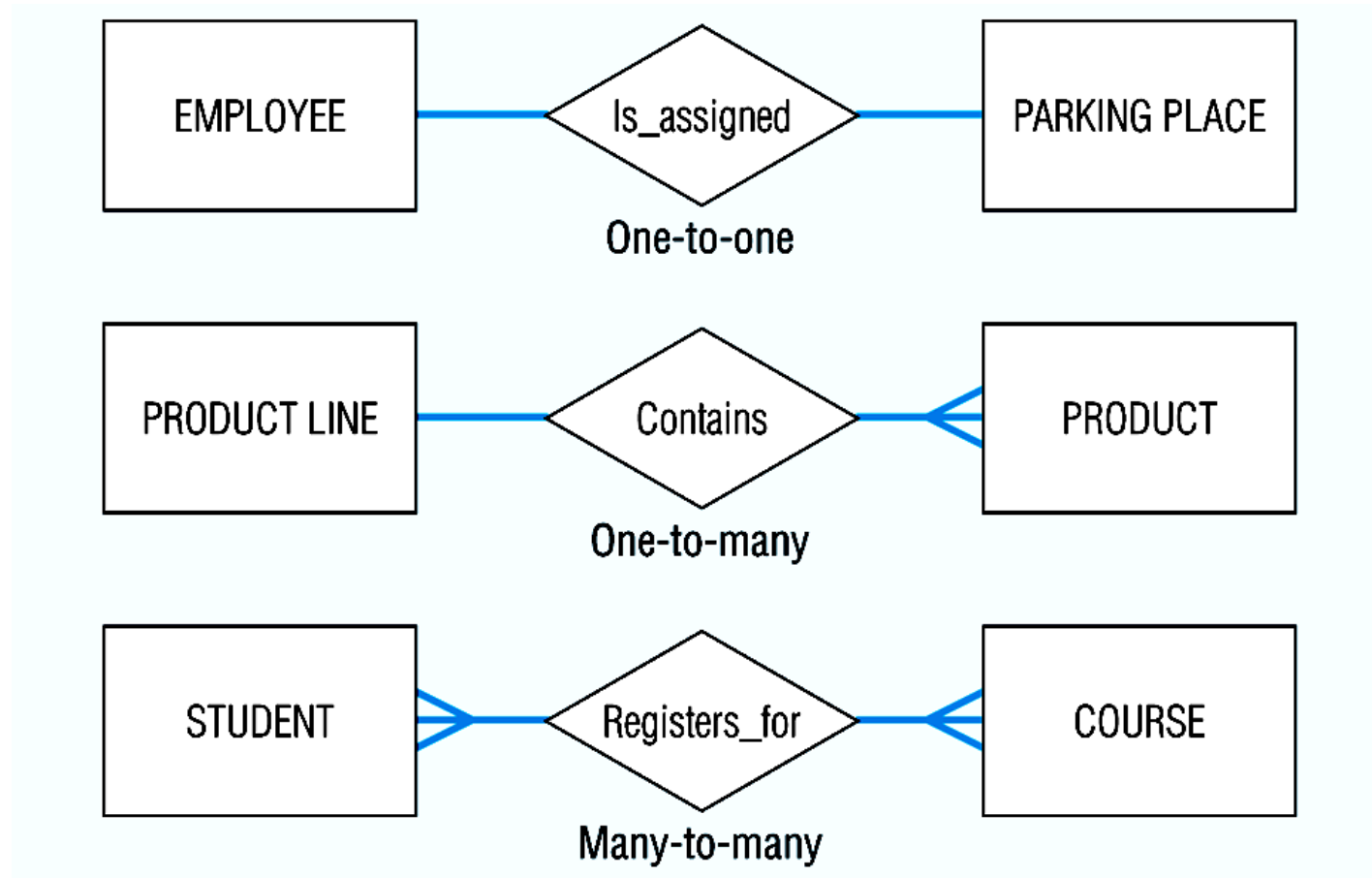
- ✓ **Many – to – Many (M:N)**

- Both entities of the relationship can have many related instances. (student – courses, employee – projects)

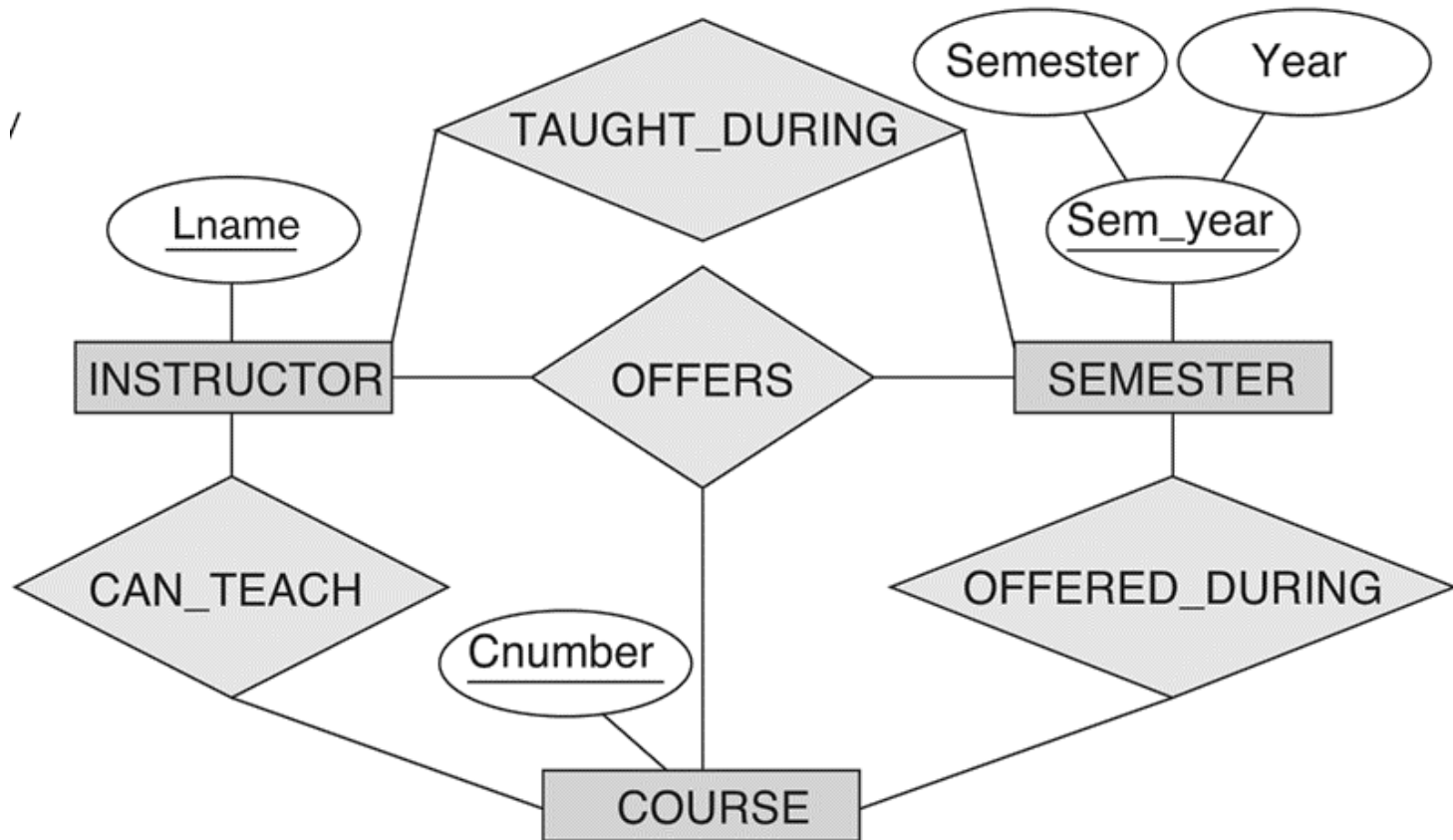
Example 1: Unary Relationship



Example 2: Binary Relationship



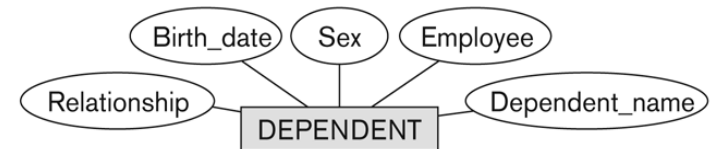
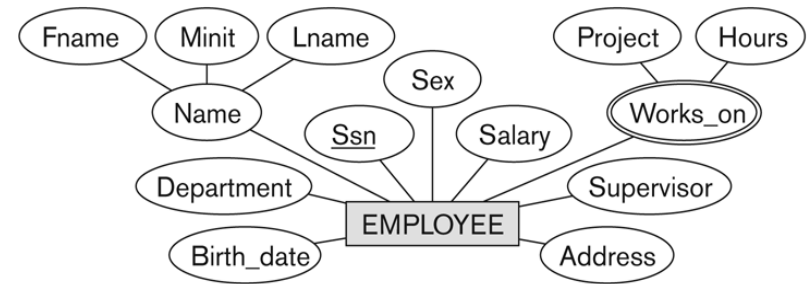
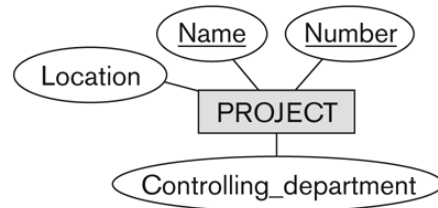
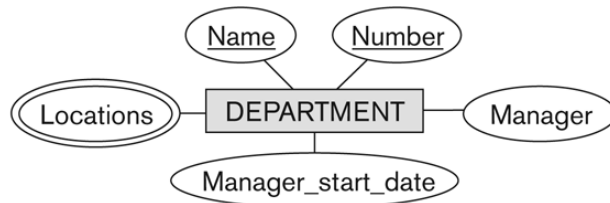
Example 3: Ternary Relationship



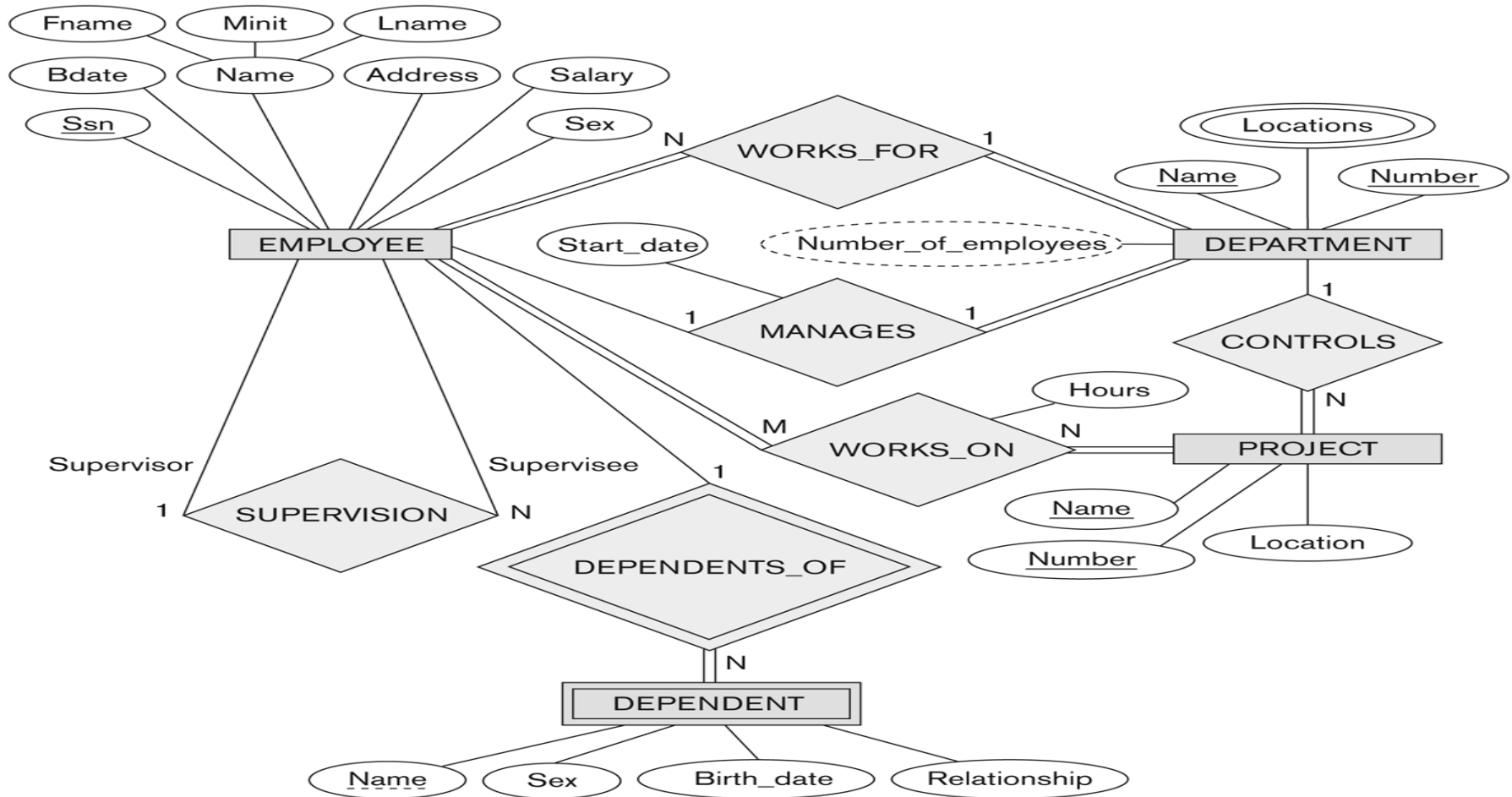
A look at the Analysis & Design Phases of Database Development

- Example of Requirements collecting of the Company:
 - The company has DEPARTMENTS.
 - Each department has a name, number & an employee who manages the department.
 - We keep the record of the starting date of manager.
 - A department may have some locations.
 - Each department controls a number of PROJECTs. Each project has a name, number and location.
 - We store each EMPLOYEE's ID, number, address, salary, gender and birthdate.
 - Each employee works for one department but may work on many projects.
 - We keep track of the number of hours per week that an employee works on each project.
 - We also keep record of the supervisor of each employee.
 - Each employee may have a number of DEPENDENTs.
 - Each dependent has name, sex, birthdate and relationship to the employee.

Conceptual Modeling



ER Design/Modeling



Questions ...?

